ShopTrack

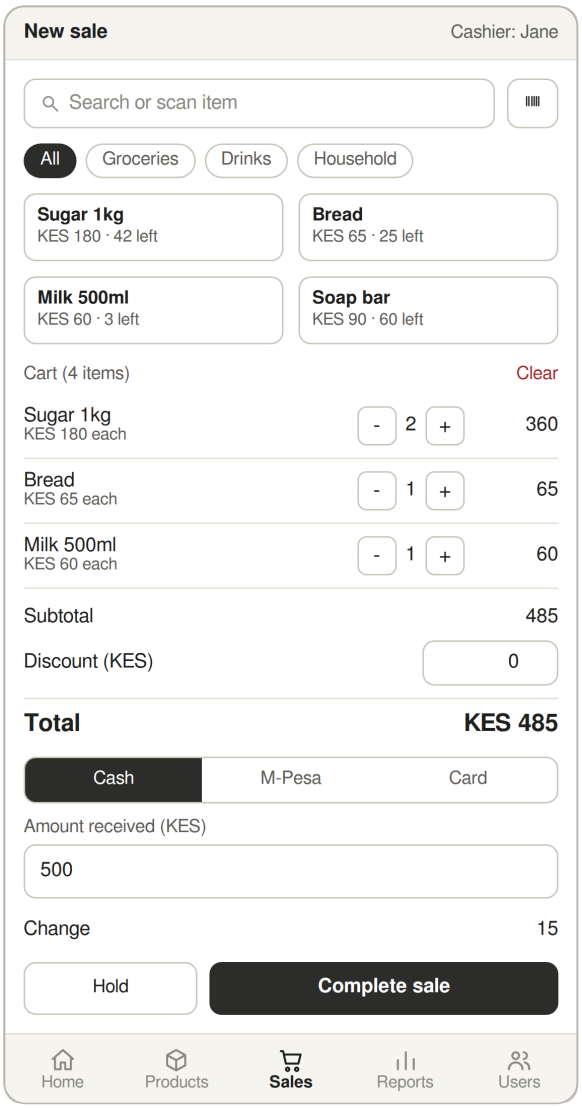
A simple Inventory and Sales System

Every duka, pharmacy, and hardware store needs to know what it has, what it sold, and what to reorder. We build a small, working system that does exactly that.

Simple · Finishable · Real problem

At a glance

Build	Python, Flask, SQLite
Core	6 steps, then it is demo-ready
Extras	Only if time allows
Screens	9 wireframes included



The sales screen: the heart of the project

> *"A working demo beats an ambitious project that never gets finished."*
>
> Our approach

The pitch

Why this project is a good fit for our group

> Let's build a simple Inventory and Sales System for a small shop. Every duka, pharmacy, and hardware store needs one, so we show a real problem and a working solution. It is small enough to finish: Python, Flask, and SQLite, with no AI or payment APIs to start. We build the basics first, and if we have time left, we add extras.

What the system does

Products
Add, edit, delete, and search items

Stock
Record deliveries, track quantities

Sales
Cart, discount, payment, auto stock deduction

Receipts
Printable receipt after every sale

Alerts
Flag items below their reorder level

Reports
Daily sales and best-selling items

Login
Admin and cashier roles

Dashboard
Today's numbers at a glance

Why it is a safe choice

Easy to understand
Everyone knows how a shop works, so nobody needs the problem explained.

Easy to demo
Add a product, sell it, and watch the stock drop. The result is visible.

Easy to extend
Extras such as charts or M-Pesa are optional, never required.

Easy to finish
A clear order of steps keeps us moving without getting stuck.

The one rule
Finish each step fully before starting the next. After step 6 we already have a complete, defensible project.

How we will build it

Three phases. We commit to Phase 1 and move at our own pace.

Phase 1: Core build

Do in order

1. **Setup and planning**
   Agree on features, create a shared GitHub repo, install Flask, and create the products, sales, and sale_items tables in SQLite.

2. **Login and roles**
   Build a login page. Admins manage everything; cashiers can only make sales.

3. **Product management**
   Forms to add, edit, delete, and list products with name, price, quantity, and reorder level.

4. **Stock updates**
   A receive-stock form that increases quantities when new goods arrive.

5. **Sales screen**
   Search a product, add it to the cart, click Complete Sale, and stock drops automatically. The heart of the project.

6. **Receipts**
   Show a printable receipt after each sale.

Core complete: we already have a demo-ready project

Phase 2: Polish and finish

Makes it stronger

7. **Low-stock alerts**
   Highlight products that fall below their reorder level.

8. **Reports**
   Show the daily sales total and best-selling items.

9. **Testing and demo**
   Load realistic sample data, fix bugs, and rehearse the presentation.

Phase 3: Optional extras

Only if time allows

Sales chart on the dashboard, expiry date tracking, M-Pesa payment, and sales forecasting. Details on the next page.

Tool stack and why

Beginner-friendly tools that we can learn and finish with

Tool	Used for	Why we chose it
Python	Core language	We already know it, so there is no new learning curve.
Flask	Web framework	Lightweight and quick to learn. Much simpler than Django for a first web project.
SQLite	Database	Built into Python with no server to install. Easy to move to PostgreSQL later.
HTML, Bootstrap, Jinja	Screens	Ready-made buttons, forms, and tables, so we do not design from scratch or need JavaScript.
ReportLab	PDF receipts	A simple library, only needed for the receipt step.
Git and GitHub	Teamwork	Everyone works in parallel without overwriting each other's code.

How the pieces fit together



Optional extras (only after step 6)

Sales chart
Chart.js or Matplotlib on the dashboard

Expiry tracking
Flag items nearing expiry, useful for pharmacies

M-Pesa payment
Daraja API with STK push

Sales forecasting
Pandas and scikit-learn suggest reorder quantities

The data we store

users
Staff, roles, and password hashes

products
Name, SKU, prices, quantity, reorder level

categories
Groups such as groceries and drinks

suppliers
Who delivers the stock

purchases
Each delivery received

purchase_items
Items and costs in a delivery

sales
Each completed sale and payment method

sale_items
Items and prices in a sale

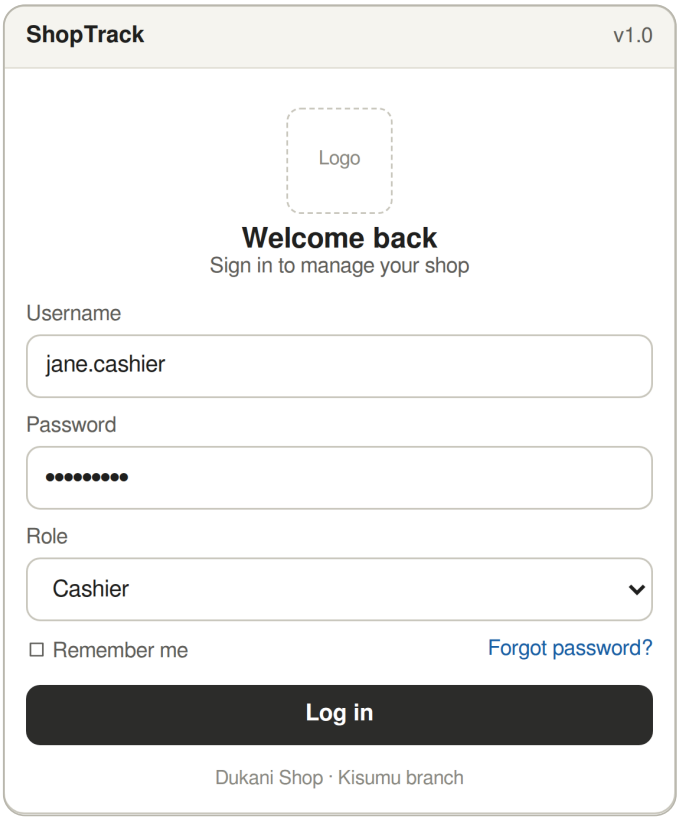
Wireframes

Screens 1 and 2 of 9

1. Login

Step 2

Staff sign in. What they can see depends on their role.



What we build

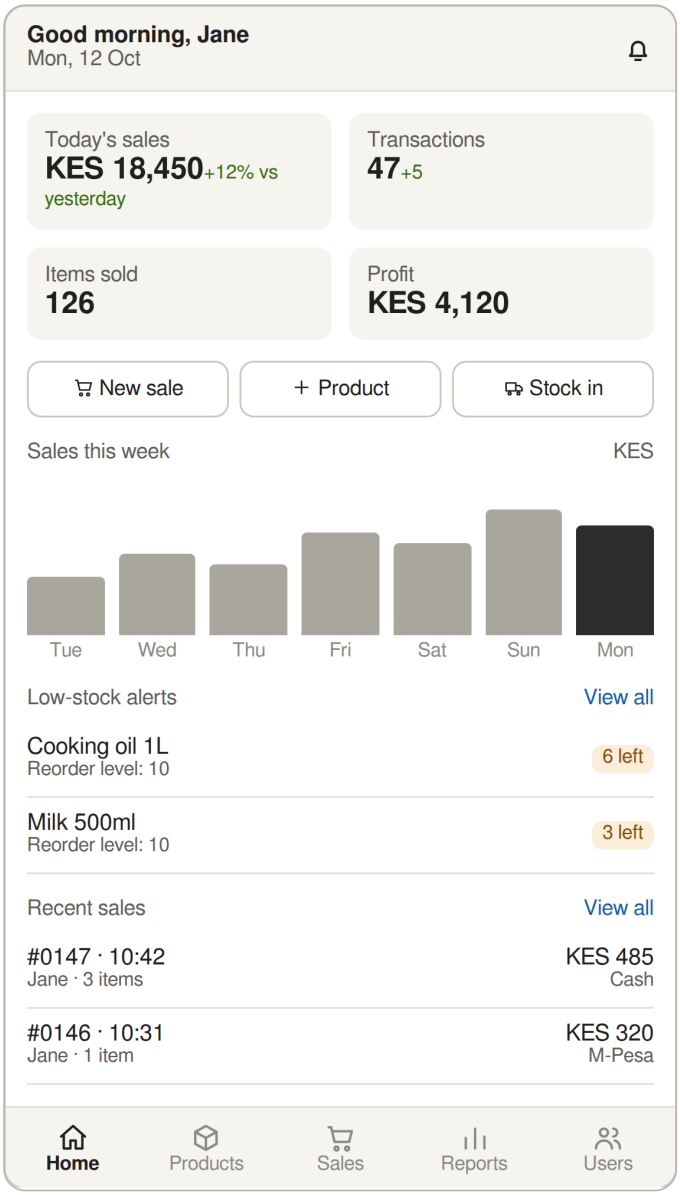
- Check username and password hash
- Send each role to its own home screen

Data: users

2. Dashboard

Steps 7 and 8

Today's sales, a weekly chart, low-stock alerts, and recent sales.



What we build

- Totals from today's sales
- Low-stock list from items below reorder level

Data: sales, sale_items, products

Wireframes are rough sketches with sample data. The final design may differ.

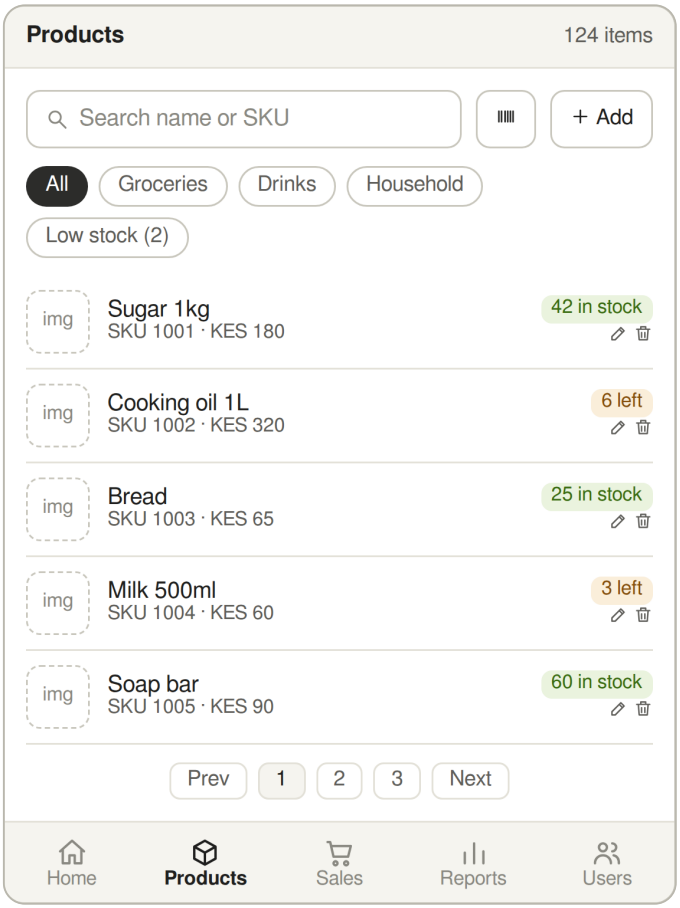
Wireframes

Screens 3 and 4 of 9

3. Products

Steps 3 and 7

Search, filter, edit, and delete items. Low stock is flagged.



What we build

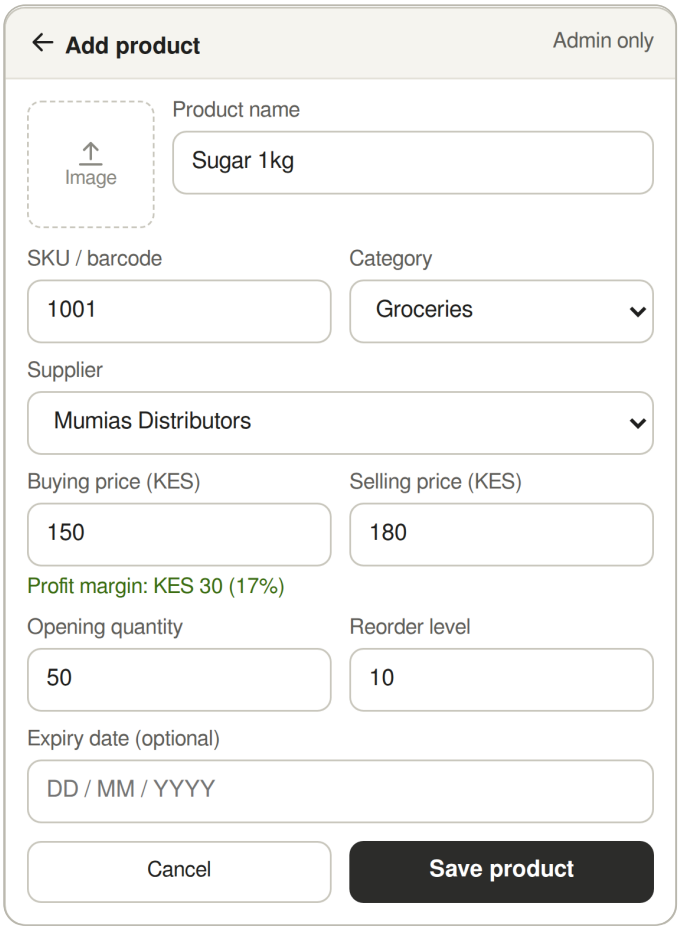
- List with search and category filter
- Edit and delete for admins only
- Badge compares quantity to reorder level

Data: products, categories

4. Add product

Step 3

One form for prices, quantity, reorder level, and optional expiry date.



What we build

- Validate prices and quantities
- Margin is selling price minus buying price

Data: products, categories, suppliers

Wireframes are rough sketches with sample data. The final design may differ.

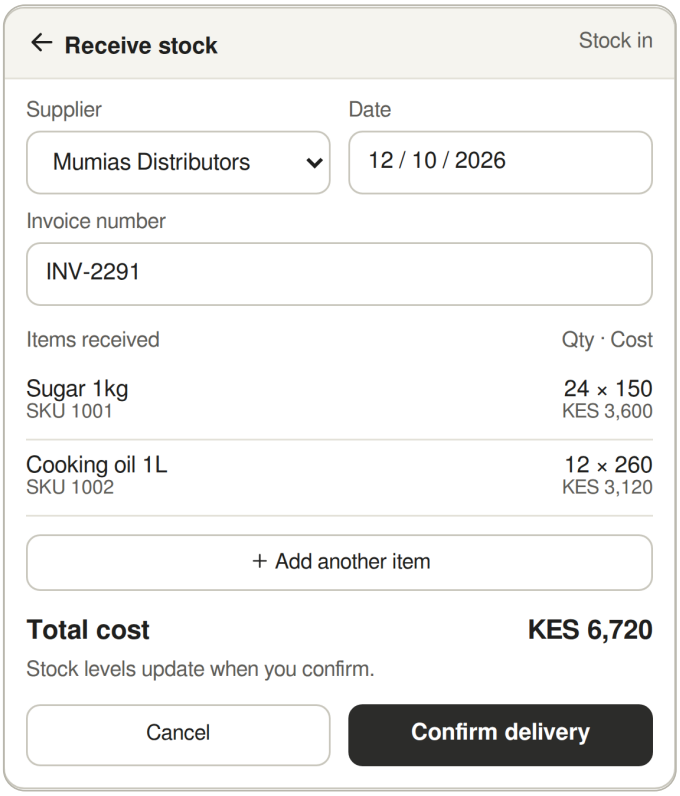
Wireframes

Screens 5 and 6 of 9

5. Stock in

Step 4

Record a supplier delivery. Stock goes up when confirmed.



What we build

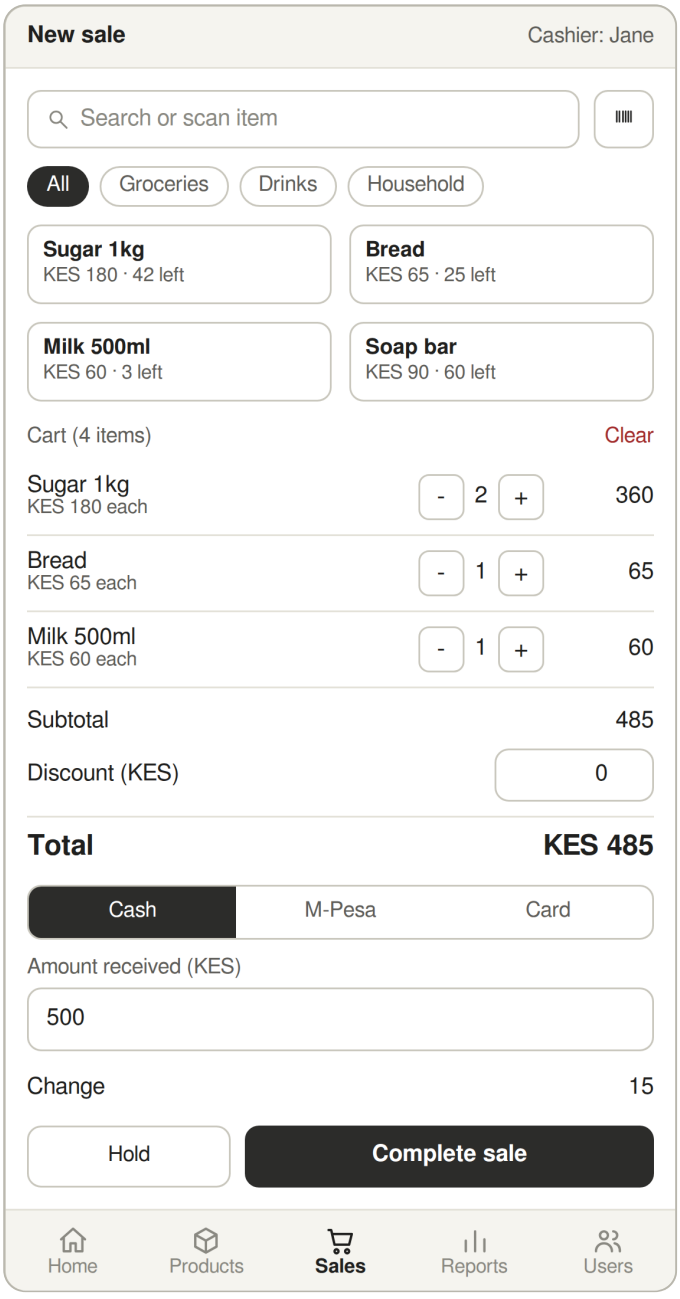
- Save the delivery and its items
- Raise product quantities on confirm

Data: purchases, purchase_items, suppliers

6. Sales

Step 5

Build a cart, take payment, and watch stock drop automatically.



What we build

- Keep the cart in the session
- On Complete Sale, save the sale and deduct stock in one database transaction

Data: sales, sale_items, products

Wireframes are rough sketches with sample data. The final design may differ.

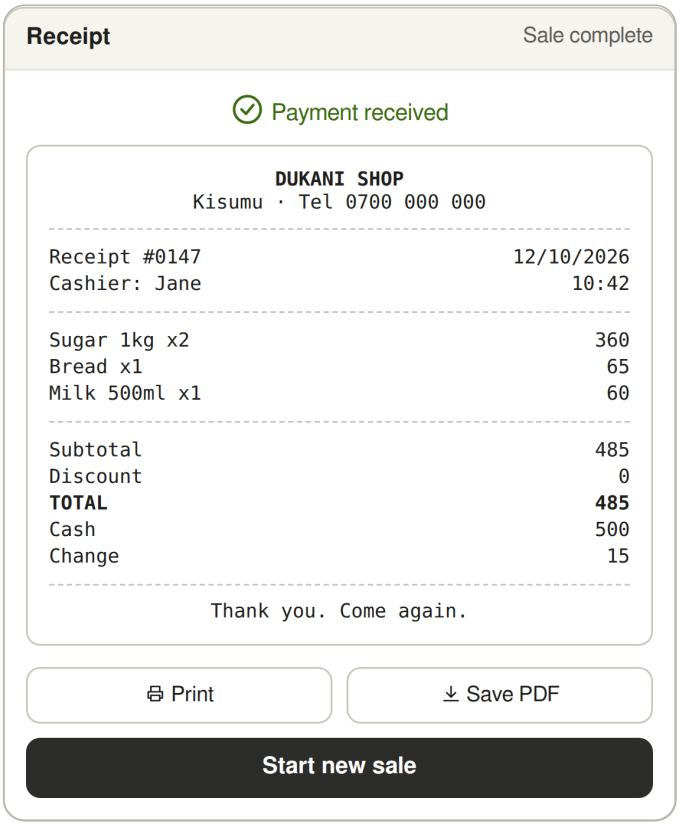
Wireframes

Screens 7 and 8 of 9

7. Receipt

Step 6

A printable receipt after every sale, with a PDF option.



What we build

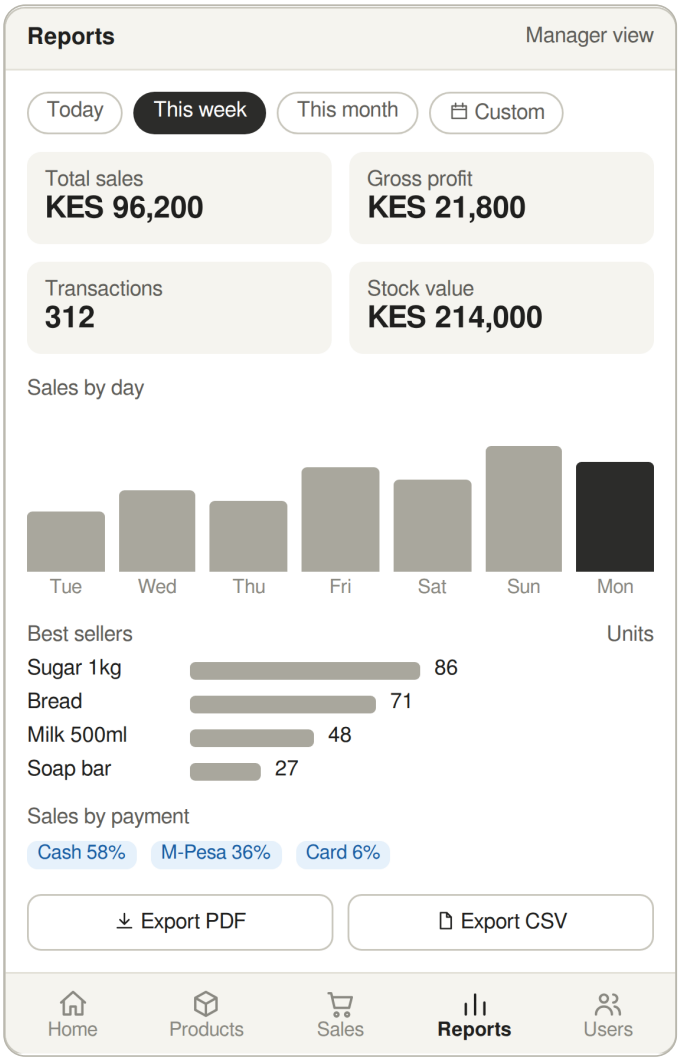
- Load the saved sale by receipt number
- Print view, plus PDF with ReportLab

Data: sales, sale_items

8. Reports

Step 8

Sales, profit, best sellers, and payment breakdown for any period.



What we build

- Group totals by day and payment method
- Best sellers from sale items
- Export to PDF or CSV

Data: sales, sale_items, products

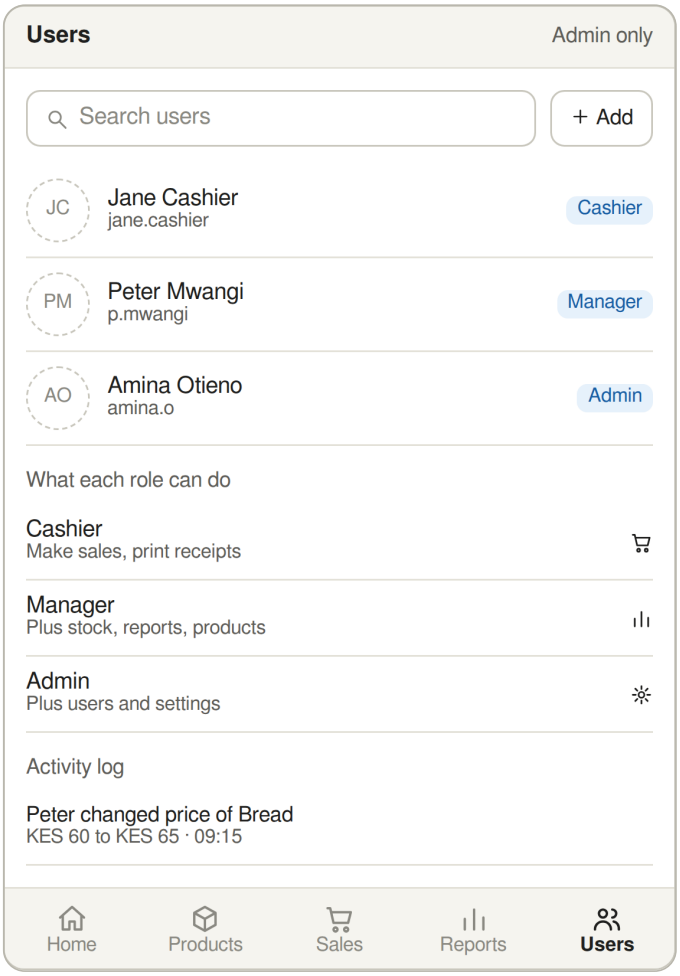
Wireframes are rough sketches with sample data. The final design may differ.

Wireframes

9. Users

Step 2

Admins manage users and roles. An activity log shows who changed what.



What we build

- Admin adds users and assigns roles
- Activity log records price and stock changes (optional extra)

Data: users

The demo story

1. Log in as a cashier.
2. Add a product and receive stock.
3. Make a sale on the sales screen.
4. Watch the stock drop.
5. Print the receipt.
6. Open the report and low-stock alerts.

Shown as optional extras

M-Pesa payment, charts, expiry dates, and the activity log appear in the wireframes but are not required for the core build.

Wireframes are rough sketches with sample data. The final design may differ.